

SEC P vs NP — notebook summary

SEC (autonomous) — attributed to Ludovico Kubler

2026-05-11 05:00 UTC

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Generated 2026-05-11 05:00 UTC

- Cycles recorded: **586**
- Time span: 415.3h (~1.41 cycles/h)
- Notebook: /home/ludo/Scrivania/SEC/research/pvsnp_notebook.jsonl

Verdict distribution

Verdict	Count
INCONCLUSIVE	549
BARRIER_HIT	18
FALSIFIED	15
SUPPORTED	4

Mathematical fields explored (field_A)

Field	Cycles
Real Algebraic Geometry	32
Representation Theory of Symmetric Groups	17
Schur-Weyl Duality	16
Free Probability	14
Matroid Theory	12
Additive Combinatorics	10
Polymatroid Theory	8
Algebraic Geometry	7
Random Matrix Theory	7
Noncommutative L^p Geometry	7
Finite Geometry	6
Spectral Graph Theory	6
Plethysm Theory	6
Ergodic Circuit Framework (communication complexity via dynamical systems)	4
{“framework_name”: “Ergodic Circuit Framework”, “math_branch”: “COMM_COMPLEXITY”}	4
Fourier Analysis on Boolean Functions	4
Fourier Analysis of Boolean Functions	4
Algebraic Geometry of Secant Varieties	4
FOURIER_ANALYSIS	4

Field	Cycles
Fourier analysis of boolean functions	3
Algebraic Topology	3
matroid theory	3
Coarse Geometric Lifting (CGL) — Geometric Complexity Theory	3
Free Probability Theory	3
{“framework_name”: “Tropical Circuit Weight Analysis (TCWA)”, “math_branch”: “BOUNDED_ARITHMETIC”}	3
Tropical Circuit Weight Analysis (BOUNDED_ARITHMETIC)	3
Schur-Weyl duality	3
Diophantine Approximation	3
Persistent Homology	3
Noncommutative Geometry	3

Last 15 cycles (chronological)

Time	Verdict	Title
2026-05-11 00:01 UTC	INCONCLUSIVE	Homotopy Group Rank Inverse Proportional to Disjointness Communic
2026-05-11 00:27 UTC	INCONCLUSIVE	Diophantine Approximation Error Inversely Proportional to Communi
2026-05-11 00:37 UTC	INCONCLUSIVE	Hypergraph Matching Complexity for Monotone DNF
2026-05-11 00:43 UTC	INCONCLUSIVE	Additive Energy Threshold for ACC ⁰ Circuit Size in CNF Formulas
2026-05-11 00:50 UTC	INCONCLUSIVE	SOS Refutation Size Bounded by Curve Genus of CSP Constraints
2026-05-11 01:05 UTC	INCONCLUSIVE	Tutte Polynomial Evaluation Bounded by Monotone k-CLIQUE Circuit
2026-05-11 01:20 UTC	INCONCLUSIVE	Young Tableau Count Exponential Gap in Disjointness Communication
2026-05-11 01:40 UTC	INCONCLUSIVE	Fourier Coefficient Sum Inverse Proportional to Disjointness Disc
2026-05-11 01:51 UTC	INCONCLUSIVE	SOS Refutation Degree Bounded by Clause Density for Random 3-SAT
2026-05-11 01:59 UTC	INCONCLUSIVE	Schatten p-Norm Lower Bound on Disjointness Communication Complex
2026-05-11 03:08 UTC	INCONCLUSIVE	Matroid Rank Lower Bound for Monotone k-CLIQUE
2026-05-11 03:39 UTC	INCONCLUSIVE	Symmetric Power Decomposition Complexity Gap for 3-SAT

Time	Verdict	Title
2026-05-11 04:26 UTC	INCONCLUSIVE	Specht Module Decomposition Exponential Gap Between Permanent and
2026-05-11 04:32 UTC	INCONCLUSIVE	Persistent Homology Gap in Read-Twice BPs for IP_2
2026-05-11 04:50 UTC	INCONCLUSIVE	Free Entropy Gap in Read-Twice BP Transition Matrices

How to read the reports

- reports/supported_findings.md — SUPPORTED conjectures with full test code. Paper-quality.
- reports/falsified_findings.md — FALSIFIED with counterexamples.
- reports/daily_YYYY-MM-DD.md — chronological log by day.
- pvsnp_notebook.jsonl — raw JSONL, one entry per cycle (source of truth).

PDF export

Install pandoc + a LaTeX engine once (sudo apt install pandoc texlive-xetex), then:

```
cd research/reports
for f in supported_findings.md falsified_findings.md notebook_summary.md; do
    pandoc "$f" -o "${f%.md}.pdf" --pdf-engine=xelatex -V geometry:margin=2cm
done
```